TS-I&II(R/S)-09-22-0985 Reg. No. [ ] B

B.Tech. Degree I & II Semester Regular/Supplementary Examina
September 2022

CS/EC/IT 19-200-0201B / CE/EE/ME/SE 19-200-0101A
COMPUTER PROGRAMMING
(2019 *Scheme*)

Time: 3 Hours Maximum Marks:

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Identify main components of a computer system and explain its working.
CO2: Develop flowchart and algorithms for computational problems.
CO3: Write the syntax of various constructs of C language.
CO4: Build efficient programs by choosing appropriate decision-making statements, loops and d structures.
CO5: Illustrate simple search and sort algorithms.
CO6: Demonstrate how to perform I/O operations in files for solving real world problems.
CO7: Design modular programs using functions for larger problems.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

			Marks	BL	CO
I.		(8 × 3 = 24)			
	(a)	Explain the function of a Linker and Loader.	3	L2	1
	(b)	Write an algorithm to print the fibonacci series up to N terms.	3	L3	2
	(c)	Distinguish between break and continue.	3	L1	3
	(d)	Explain enumerated datatypes in C.	3	L1	3
	(e)	Compare iteration and recursion.	3	L2	4
	(f)	Describe fseek(), ftell() and rewind() functions.	3	L2	6
	(g)	Write a function to find the GCD of two positive integer numbers.	3	L3	7
	(h)	Explain function definition and declaration with its syntax.	3	L2	7

PART B
(4 × 12 = 48)

II.	(a)	Describe different types of operators that are included in C.	8	L2	3
	(b)	Explain the function of compilers and interpreters.	4	L1	1
		OR			
III.	(a)	Explain problem solving methodology.	8	L1	1
	(b)	Discuss formatted input and output functions in C.	4	L2	3
IV.	(a)	Explain the selection statements in C.	6	L1	3
	(b)	Write a C program to find the sum of digits of a given number.	6	L3	4
		OR			
V.	(a)	Discuss keywords, identifiers and constants in C.	6	L1	3
	(b)	Write a C program to calculate the sum of prime numbers in a given range.	6	L3	4

5. Describe formatted input and output functions ?
6. Write a program to print the prime numbers between 0 and 100 ?
7. Differentiate between call by value and call by reference ?

Module 3

1. Write the syntax for declaring a structure ?

2. How are functions declared?

3. Compare Linear searching and Binary searching.

4. Write a C program to search for an element in an array using Binary searching Algorithm.
5. Write a program to reverse a string
6. Explain the user-defined data types ?
7. Write a program to perform linear search in an array ?
8. What is bubble sorting? Write a program to sort a list of names in alphabetic order using bubble sorting

Module 4

1. What are command line arguments ?

2. Explain 'fseek' with an example program ?

3. Explain all file handling operations ?

4. Write a program to open a file and change all characters to lower case ?

5. Compare array of pointers and pointer to an array ?

6. What is pointer? Write the syntax for declaring a pointer ?

7. Write a C program to copy content of one file to another file?

PREVIOUS YEAR QUESTIONS

Module 1

1. Difference between compiler and interpreter?

2. What is application software and system software? Find examples for each?

3. Draw the block diagram of a digital computer. Describe each part ?

4. Explain the steps in execution of a program ?

5. Draw the flow chart for checking a number is odd or even ?

6. What is system software?

Module 2

1. Explain the different control statements in 'C' ?

```
int main()

int a=10,b=6;
int c=a*b++;
   printf("%d %d %d",
         a,b,c); int d =
         a*++b;

printf("%d%d %d",
a,b,d);
```

2. Predict the output of the following code ?

3. Write a program to find the GCD of 2 numbers ?

```
  void main()

int i = 1, j = 2;

printf("%d %d",
++i+j++,i--);
```

4. Predict the output of the following code.

			Marks	BL	CO	PO
VI.	(a)	Write a C program to read two strings from the keyboard and concatenate them without using built-in function.	6	L3	4	1,3,4
	(b)	Write a program to sort a list of numbers in descending order.	6	L3	5	1,3,4
		OR				
VII.	(a)	Write a C program that will read a word and rewrite it in alphabetical order.	6	L3	4	1,3,4
	(b)	Write a C program to perform sequential search and binary search using functions.	6	L3	5	1,3,4
VIII.	(a)	Write a C program for array multiplication using pointers.	6	L3	4	1,2,3
	(b)	Write a file copy program in C that copies a file into another.	6	L3	6	1,2,3
		OR				
IX.	(a)	Write a C program to create a structure that contains roll number, name and marks scored in 6 subjects in a class and prepare a mark list based on their total marks.	6	L3	4	1,2,3
	(b)	Explain dynamic memory allocation in C.	6	L2	6	1,2,3

Bloom's Taxonomy Levels
L1 = 25%, L2 = 29.16%, L3 = 45.83%

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8. Differentiate malloc and calloc.
9. What are pointers?
10. Define union.
11. What is a structure? Differentiate 'array of structures' and 'arrays within structures' with suitable examples.
12. Write a program using structure to read and display details of 'n' employees. The details should include employee name, age, phone number and salary. The net salary is calculated as salary=basic pay+ allowance- deductions.
13. Write a program to separate Odd and Even numbers in a file and explain the program.

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17. Explain any two variants of if statement.
18. Differentiate break and continue with the help of a program.
19. Write a program to print prime numbers upto 'n'.
20. Explain the working of do…while loop with an eg.
21. Write a program to sort an array using function.
22. What are operators in 'C' programming? Explain.
23. Discuss about looping statements in C with example?
24. State the differences between break, continue and exit().
25. Write a program to count the number of vowels in a given string.
26. Explain switch statement with syntax.
27. A cloth showroom has announced the following seasonal discounts on purchase of items:

Purchase Amount	Discount	
	Mill Cloth	Handloom Items
0-100	-	5%
101-200	5%	7.5%
201-300	7.5%	10.0%
Above 300	10.0%	15.0%

Write a program using **switch** and **if** statements to compute the net amount to be paid by a customer.

MODULE III:

1 What is a function prototype?
2 What is the need for functions in C?
3What do you meant by scope of a function?
4 Differentiate actual and formal arguments.
5 What is the use of exit() function?
6 What is recursion? Explain with an example.
7 Explain goto statement with example.
8 Write a program to find transpose of a given square matrix.
9 Explain briefly the concept of arrays in C. Write a program to sort an array of n elements.
10 Write a program to implement matrix multiplication.

MODULE IV:

1. How do you access a file randomly in C?
2. What are file inclusion directives?
3. Define File pointer.
4. Write a C program to display address and contents of different types of variables using pointers.
5. Discuss about the concept of array of pointers.
6. Write a program to print details of 50 students using structures in 'C' programming.
7. Write a 'C' program to read the content of file using 'fgets'.

QUESTIONS

MODULE I:

1. What is an algorithm?
2. What is a computer program?
3. Describe the features of C.
4. Differentiate top down and bottom up system design techniques.
5. What do you meant by modular programming?
6. Discuss about the 2 approaches in building a software.
7. Explain flowchart and the various symbols used in flowchart. Draw a flowchart to simulate a simple calculator.
8. Write an algorithm to find the biggest among 3 numbers.
9. Explain the classifications of algorithm.
10. Explain the classification of Programming Languages. What are virtual threats?
11. Write a short note on Assemblers, Compilers and Interpreters.

MODULE II:

1. What is a constant?
2. List decision making statements in C.
3. List atleast 4 keywords used in C.
4. What does **? :** operator stands for? Explain.
5. Differentiate "=" and "==" in C with example.
6. Discuss arithmetic operators and logical operators in C with example.
7. Write the output of the following code.

```
main()
{
  int x, y=5 ,m ,n=10;
  x=y++;
  m=--n;
  printf("X=%d, Y=%d", x,y);
  printf("M=%d, N=%d", m,n);
  }
```

8. What is the meaning of the below statement
   `for(i=0;i<1000;i++);`
9. What do you meant by a type modifier in C?
10. What is the difference between entry controlled and exit controlled loop?
11. What is a jumping statement? List any 2 jumping statements in C.
12. Differentiate type conversion and type casting in C with eg.
13. Give 2 example functions for Formatted I/O and Unformatted I/O each.
14. Explain the different types of arrays
15. Differentiate ++x and x++.
16. Write and explain the syntax of switch statement.

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ENGINEERING

FIRST MODULE TEST APRIL 2021

19-200-1201B COMPUTER PROGRAMMING

S2 CS A (2020 ADMISSION)

TIME DURATION -1HR MAXIMUM MARKS:20

Course outcomes:
On completion of the course students will be able to

CO1: Identify main components of a computer system and explain its working.
CO2: Develop flowchart and algorithms for computational problems

ANSWER ALL QUESTIONS

1.Draw the block diagram of a digital computer and explain the working of all of its units in detail. CO1 5MARKS

2.Explain the steps in execution of a program with a neat diagram. CO1 5MARKS

3.Write the algorithm and draw flowchart of finding whether the given number is palindrome or not. (eg:121)

CO2 10MARKS

OR

4.Write the algorithm and draw flowchart of finding Fibonacci series up to N terms

(eg.01123….)

CO2 10MARKS

SCHOOL OF ENGINEERING, CUSAT

B.Tech degree Second Internal Examination, July 2020

Semester II Course Title:19-200-0201B-CS/EC/IT COMPUTER PROGRAMMING

MAX. MARKS: 40 **TIME: 1 HOURS**

Part A (Answer Any Four questions) **(10 x 4 = 40)**

1. a. Write a program to check whether a given string is palindrome or not without using any string manipulation functions.

b. Write a program to find Fibonacci seris up to n terms using recursion.

c. Write a program to print details of N students in ascending order of their total marks. Use structure to create student records with name, roll no and marks.

d. Write a program to add to two integer arrays and save the result into another array using pointers.

e. Write a program to copy one file to another.

SCHOOL OF ENGINEERING, CUSAT

B.Tech degree First Internal Examination, JUNE 2020

Semester II Course Title:19-200-0201B-CS/EC/IT COMPUTER PROGRAMMING

MAX. MARKS: 40 **TIME: 1 HOURS**

Answer any FOUR questions **(10 x 4 = 40)**

a. Explain different types of programming languages? 5

b. Explain the steps involved in problem solving methodology? 5

a. Write the algorithm to check whether the given number is even or odd. Also draw the flow chart for the same. 10

a. Write any five types of operators used in C with suitable examples. 5

b. Explain different data types used in C programming language. 5

a. Write a program to check whether the given number is armstrong number or not (hint: $1^3+5^3+3^3=153$) 10

a. Using for loop, write a program to print the fibonacci series upto given number of terms. 10

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Assessment	B.TECH. DEGREE INTERNAL ASSESSMENT (SERIES TEST – I)	Semester	II
Course Code and Name	19-200-0201B -CS/IT/EC COMPUTER PROGRAMMING	Division	IT/CS/EC
Date of Exam	29/07/2022(FN) Max. Marks : **40**	Duration	2Hrs

CO1	Identify main components of a computer system and explain it working.
CO2	Develop flowchart and algorithms for computational problems.
CO3	Write the syntax of various constructs of C language.
CO4	Build efficient programs by choosing appropriate decision making statements, loops and data structures.

Q.No	PART A (Answer all questions)	Marks
1.	Describe about different types of software with example.	2(CO1)
2.	Explain a general structure of C program with an example.	2(CO3)
3.	Draw the flowchart for the C program to find the largest among three numbers.	2(CO2)

4.	What will be the output of following program?Justify your answer.
	`#include <stdio.h>`
	`void main()`
	`{`
	`int x=6,y,z;`
	`y=--x;`
	`z=x--;`
	`printf("%d%d%d",x,y,z);`
	`}`
	x=6 y=5 z=5 x=4
	4,5,5
		2(CO3)

5.	Write a C program to find whether the number is positive or negative using conditional operator. e1 ? exp2 : exp1	2(CO4)

Q.No	PART B (Answer any three questions)	Marks
1.	Draw the block diagram of a computer and describe all the functional units.	10(CO1)
2.	List out and steps involved in problem solving methodology.	10(CO1)
3.	Write a c program to read a number and check whether the number is odd or even. Also check the following conditions: (using a switch statement) Case1: if the number is even then print the square of the number Case2: if the number is odd then print the cube of the number.	10(CO4)
4.	Write an algorithm and Develop a C program to print Fibonacci series of n terms.	10(CO2,CO4)
5.	Design and develop a C program to reverse a given integer and check whether it is PALINDROME or NOT.	10(CO4)